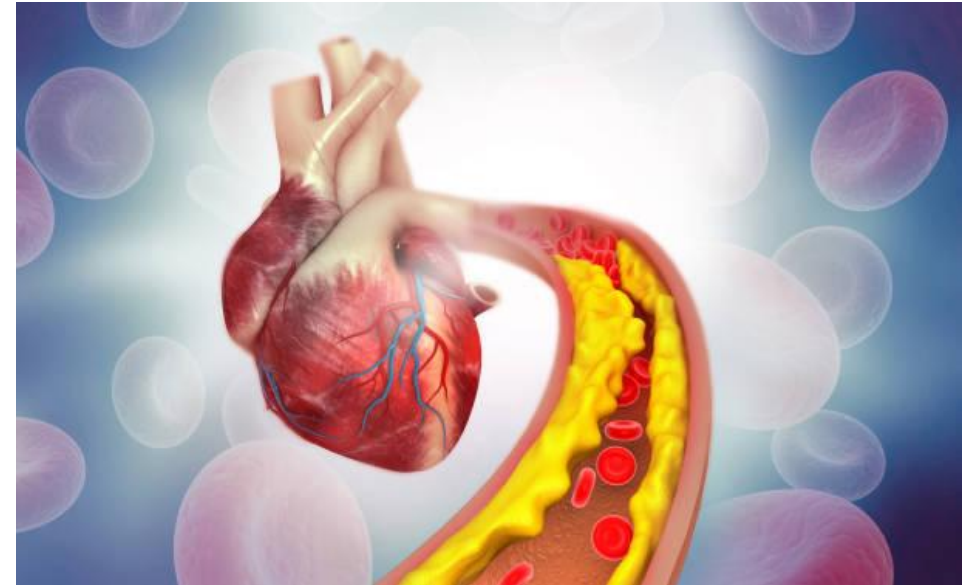


Fats in Human Health and Diseases



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NTN 1232 Fundamentals of Human Nutrition

Learning objectives

- Categorize fats and identify the composition
- List different fatty acids
- Compare the compositional differences of fats & oils
- Dietary sources of fats
- Describe the role of fats in the body
- Explain the harmful effects of fats to human health

Introduction

- A diverse group of compounds including fatty acids, triglycerides, phospholipids & sterols
- Belongs to the large group of lipids
- Can be extracted by organic solvents
- Provide essential fatty acids (EFA) and fat soluble vitamin to the diet
- Atwater factor : 9 kcal (1 g of fat gives 9 kcal)



Classification of fats

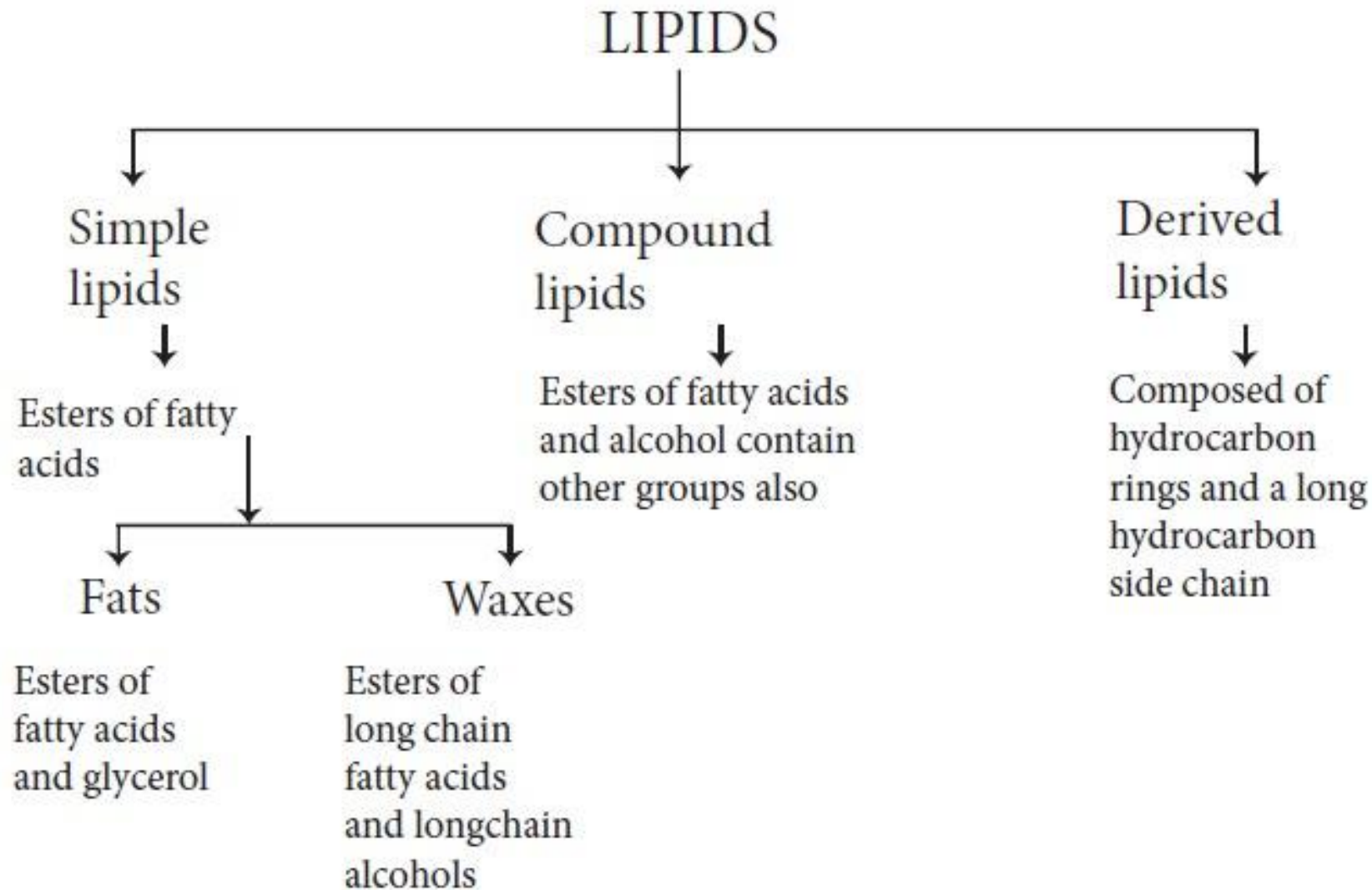
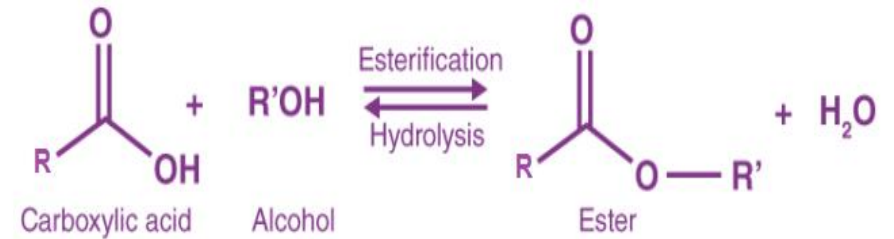


Fig 10.13: Classification of fats based on fatty acids

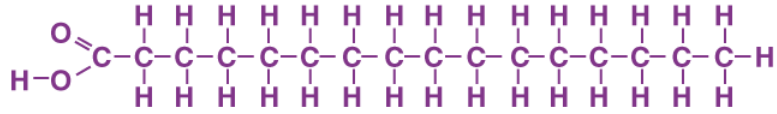
Components in Fats



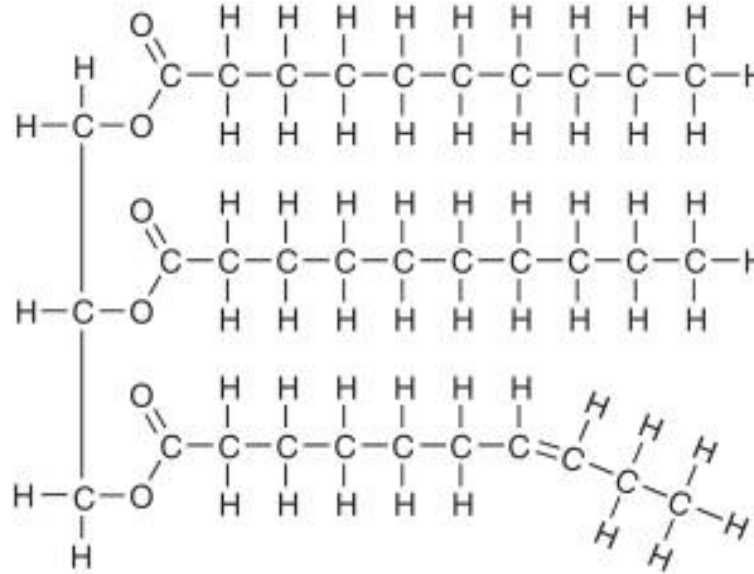
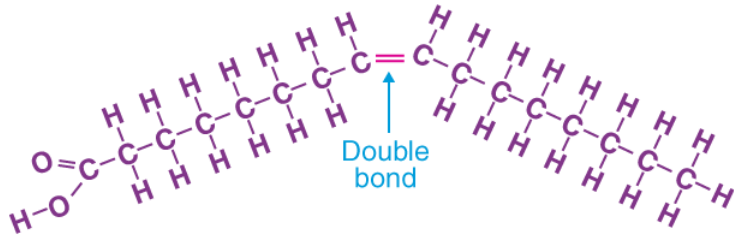
- Fatty acids : building blocks of fats
- Triacylglycerols (triglycerides) are the main dietary lipids which are stored in the body.
- Esters of fatty acids (both saturated and unsaturated) with glycerol
- Phospholipids : Contain phosphoric acid as a monoester or diesters.
- Glycolipids : Contain monosaccharide residues joined through a glycosyl linkage to a lipid
- Cholesterol
- Cholesterol ester and phospholipids are important in the structure of cell membranes, mitochondria, lysosomes and the endoplasmic reticulum.
- Lipoproteins : Made out of proteins and fats. Carry cholesterol to the cells



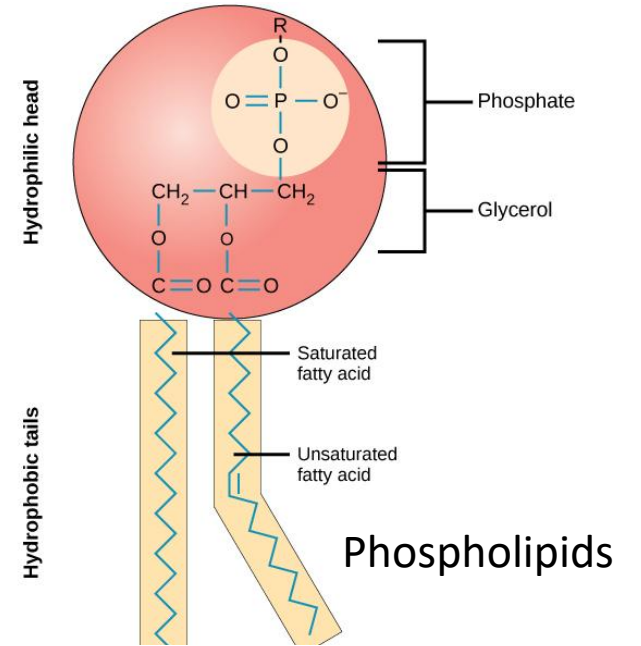
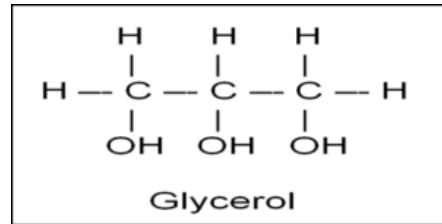
Saturated Fatty Acid



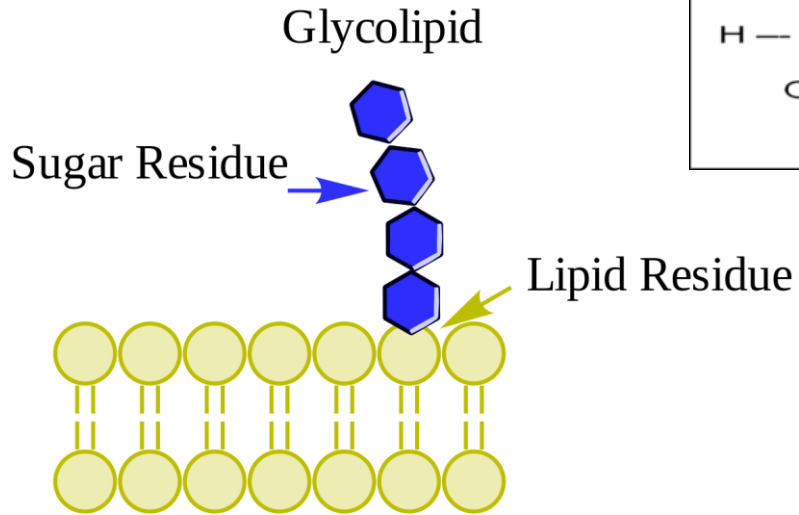
Unsaturated Fatty Acid



Triglyceride

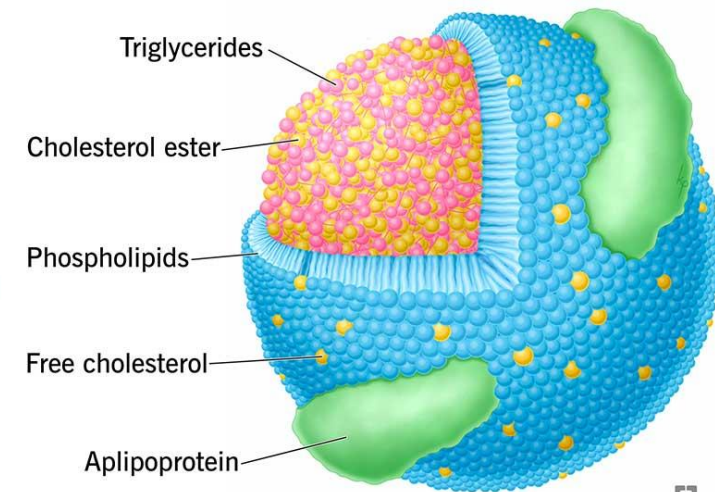
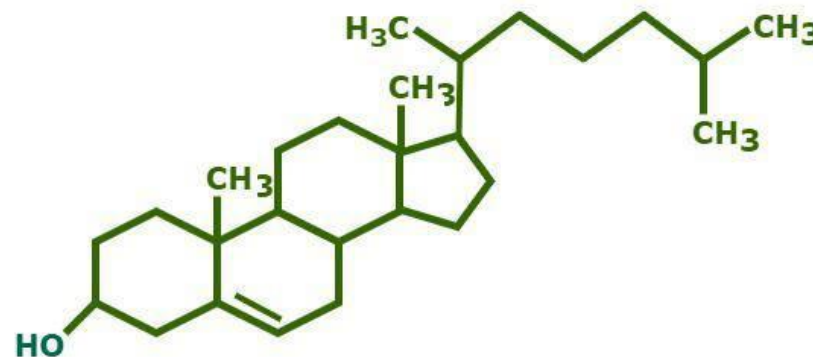


Lipoprotein



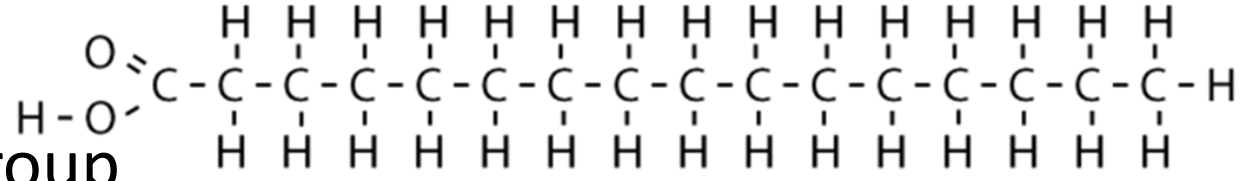
Lipid Membrane

Structure of Cholesterol



Fatty acids (FA)

- Made up of C, H, O
- Consisted of methyl group and acid group
- Number of C atoms and double bonds are vary



Ex : Butyric acid (C4:0), Palmitic acid (C16:0) and Arachidonic acid (C20:0)

Short-chain fatty acids (SCFA)

- About 5 carbon atoms
- Shorter the chain length, increase the solubility in water
- Easily break down during digestion
- Oxidize easily
- Easier the absorption

<https://www.youtube.com/watch?v=brs2nMubr84>

<https://www.youtube.com/watch?v=ebScOnAJdu0>

Fatty acids (FA)

Medium-chain fatty acids (MCFA)

- 6 to 12 C atoms in a chain

Long-chain fatty acids (LCFA)

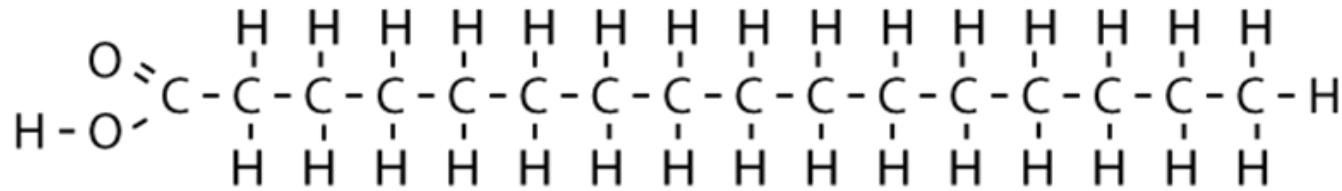
- 13 to 21 carbon atom

Very long chain fatty acids (VLCFA)

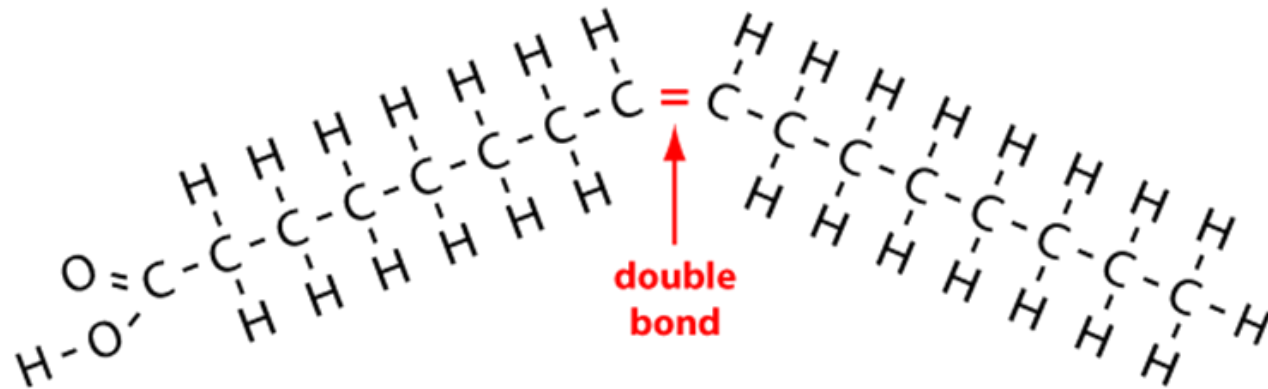
- More than 22 carbon atoms.
- The majority of Fas (both in the diet and in the body) are 16-18 carbon atom molecules

Fatty acids (FA)

saturated fatty acid

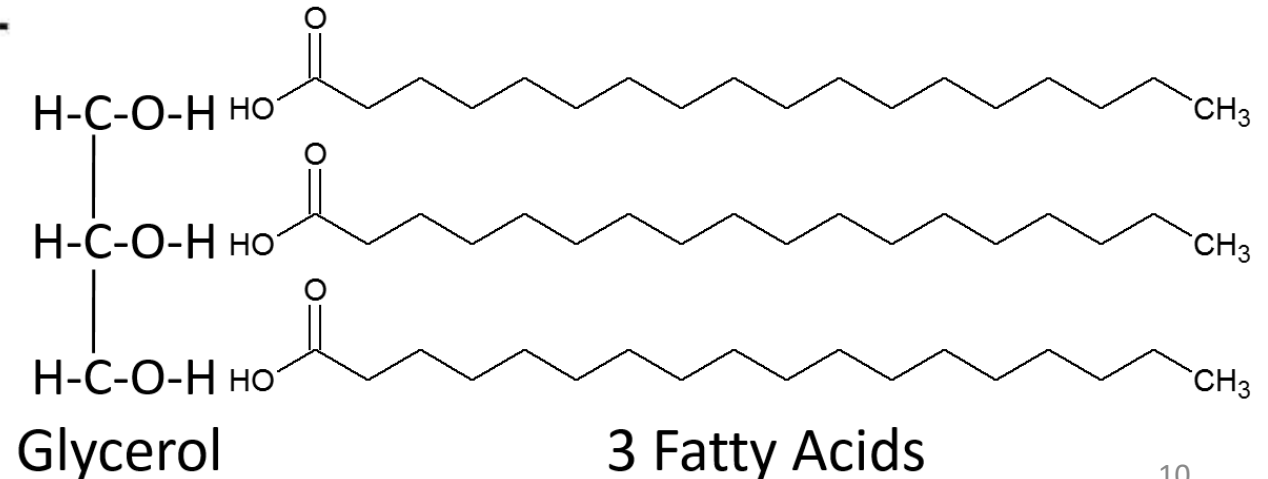
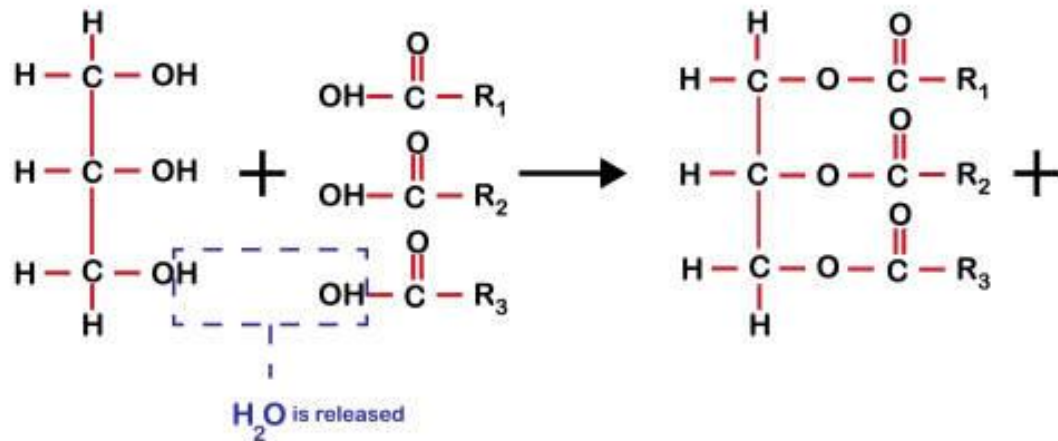
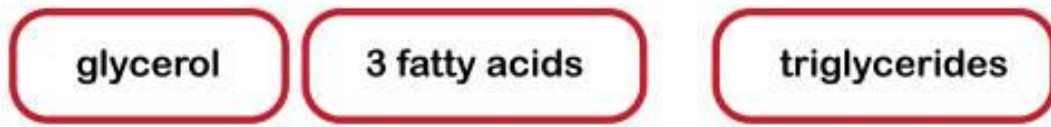
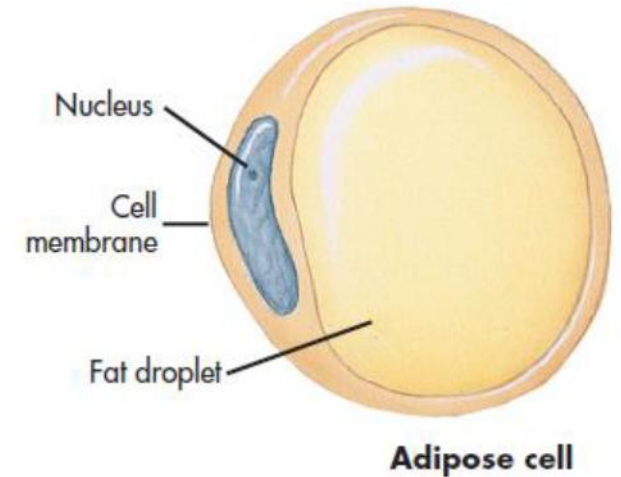


unsaturated fatty acid



Triglycerides

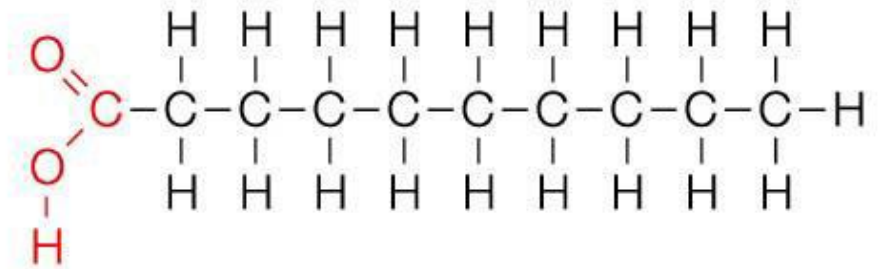
- 90% of fats in human body
- Stored form of fats in the body
- The ester derived from glycerol and three fatty acids



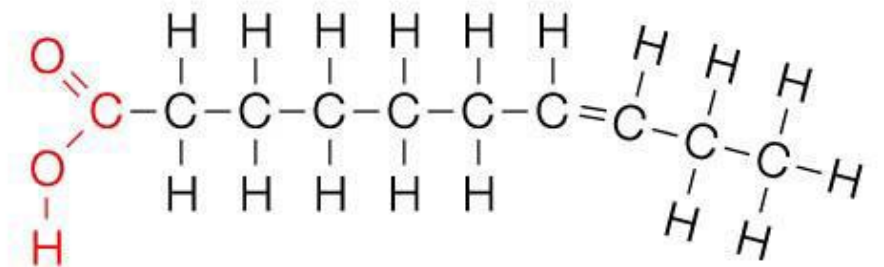
FA classifications : Saturation

- No carbon-carbon double bonds in FA chain – Saturated
- Double bonds between carbon atoms - Unsaturated

Saturated

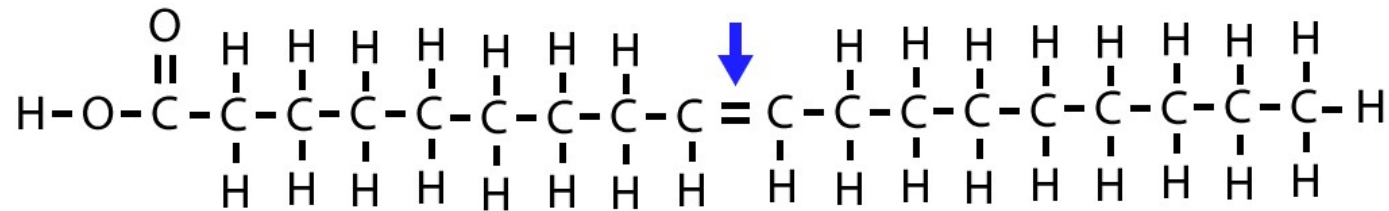


Unsaturated

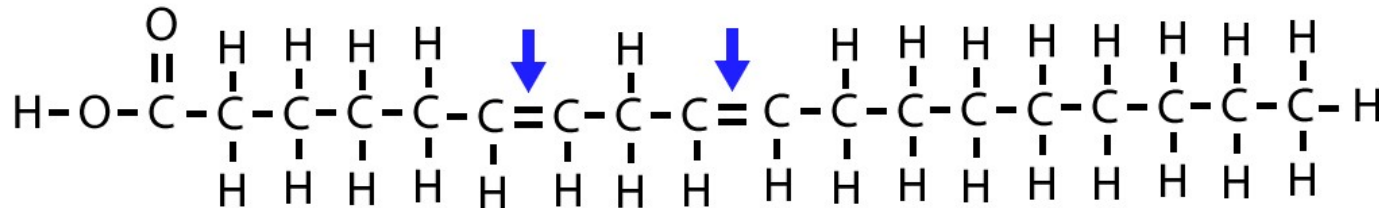


Monounsaturated (MUFA) and Polyunsaturated fatty acids (PUFA)

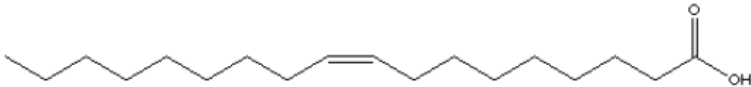
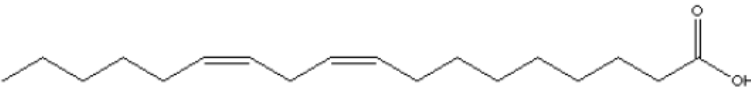
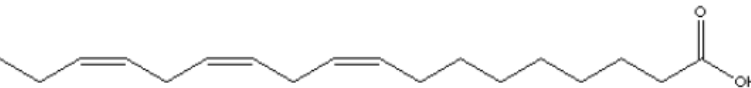
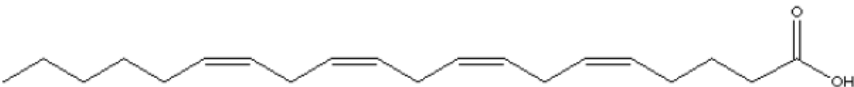
Oleic acid - Monounsaturated fatty acid



Linoleic acid - Polyunsaturated fatty acid



Unsaturated Fatty acids

Trivial name	Symbolic expression	Chemical structure
oleic acid	18:1 ω -9	
linoleic acid	18:2 ω -6	
α -linolenic acid	18:3 ω -3	
arachidonic acid	20:4 ω -6	
eicosapentaenoic acid (not trivial name)	20:5 ω -3	
docosahexaenoic acid (not trivial name)	22:6 ω -3	

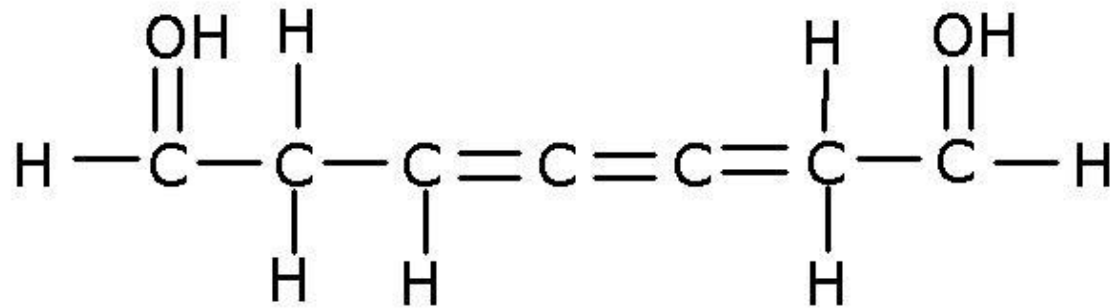
MUFA

- One double bond in the fatty acid chain
- Liquids at room temperature and semisolid or solid when refrigerated
- Plant-based oils such as olive oil, canola oil, peanut oil, safflower oil, and sesame oil.
- Nuts such as cashews, almonds, and hazelnuts



PUFA

- Contain more than one double bonds
- Essential fatty acids are included
- Corn, soybean, sunflower, safflower oil



Polyunsaturated fatty acid



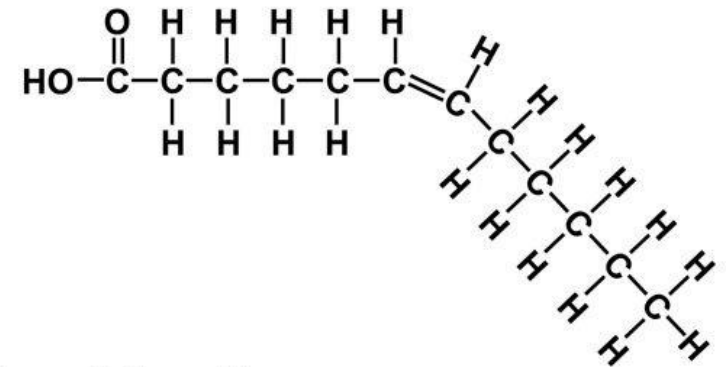
Saturated Fatty acids

Systematic Name	Trivial name
dodecanoic acid	lauric acid
tetradecanoic acid	myristic acid
hexadecanoic acid	palmitic acid
octadecanoic acid	stearic acid
eicosanoic acid	arachidic acid
docosanoic acid	behenic acid

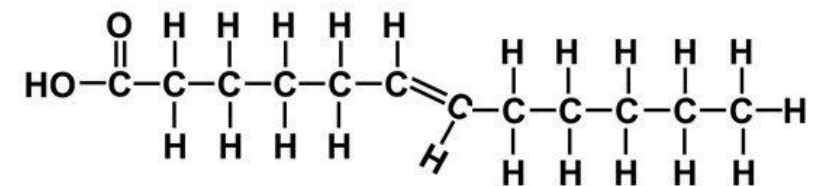
Cis and Trans FAs

- Cis fatty acids : two hydrogen atoms attached to the double bond in the same side of the carbon chain
- Trans fatty acids : two hydrogen atoms bonded to the double bond in the opposite sides of the carbon chain
- Cis fats are the typical form of unsaturated fat found in nature, while trans fats are made through hydrogenation.

cis-fatty acid

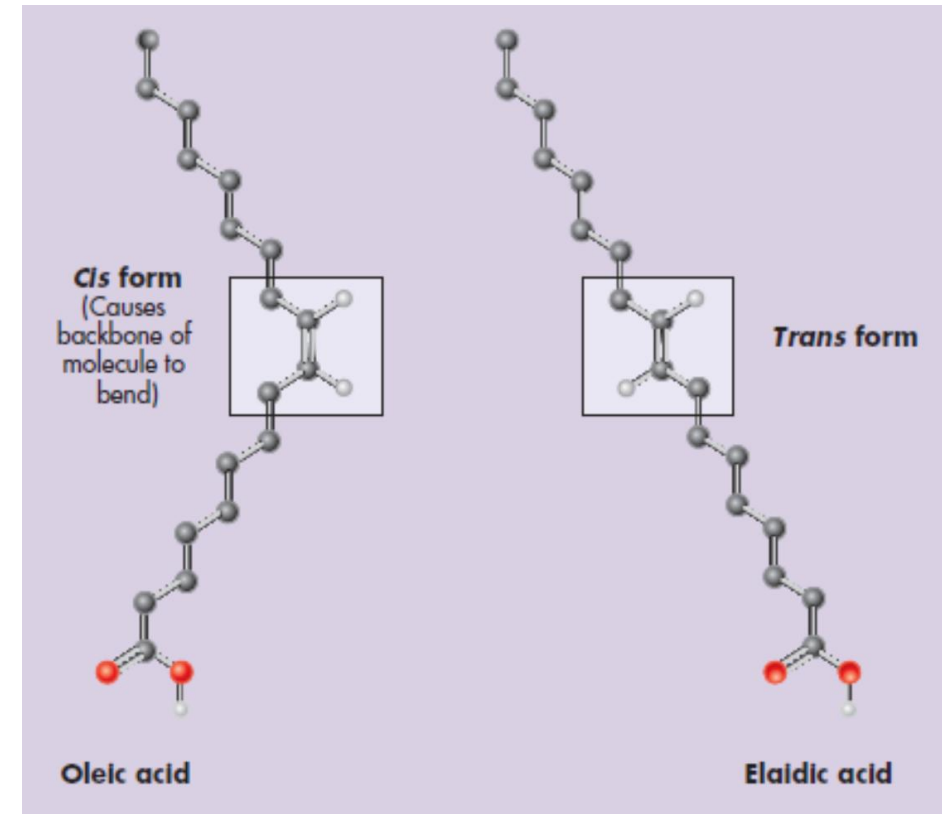


trans-fatty acid



Cis and Trans FAs

- Naturally MUFA & PUFA in cis form
- During processing some hydrogen transfer to opposite side of C-C double bond creating trans form
- Cis bond causes FA chain to bend
- Trans bond create straight chains similar to SFAs



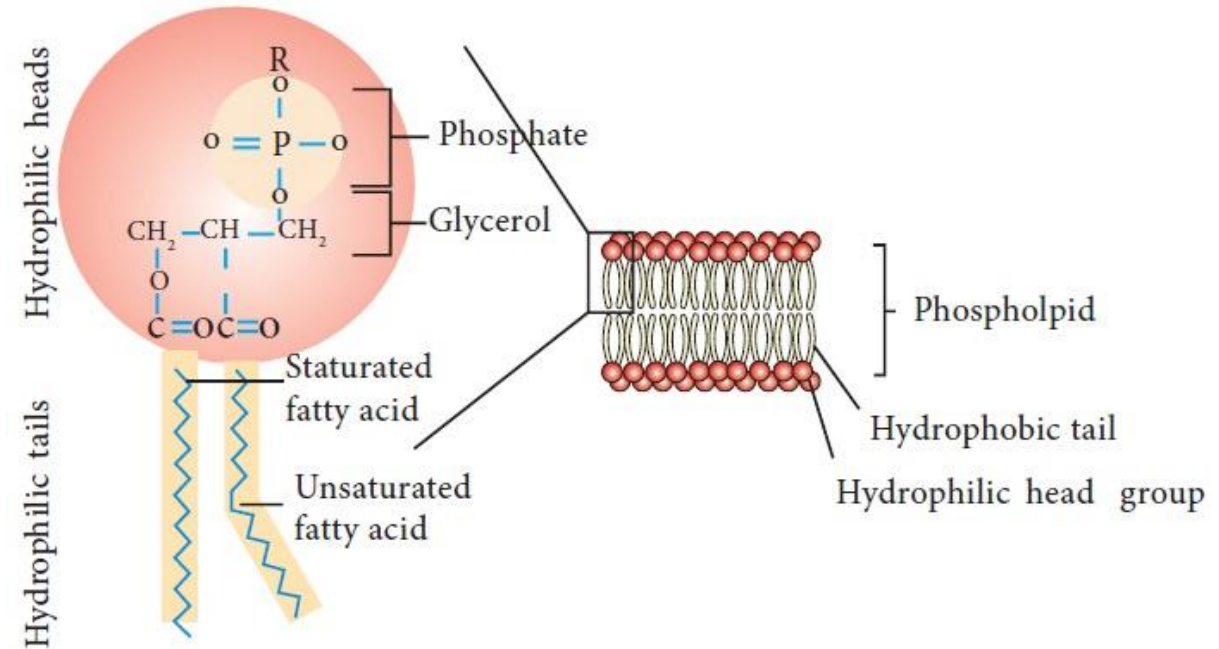
Fatty acid	Carbon chain length	Fatty acid	Carbon chain length
Saturated fatty acids		Monounsaturated fatty acids (MUFA)	
Caprylic acid	8:0	Myristoleic acid	14:1
Capric acid	10:0	Palmitoleic acid	16:1
Lauric acid	12:0	Oleic acid	18:1
Myristic acid	14:0	Gadoleic acid	20:1
Palmitic acid	16:0	Erucic acid	22:1
Stearic acid	18:0	Nervonic acid	24:1
Arachidic acid	20:0	Polyunsaturated fatty acids (PUFA)	
Behenic acid	22:0	Linoleic acid	18:2
Lignoceric acid	24:0	Linolenic acid	18:3
		Eicosadienoic acid	20:2

It is the chemical and physical nature of the constituent fatty acids that decide those of TAG (de Man 1998; Orsavova et al. 2015)

No. of Carbons	Fatty Acid	Structure	
2	Acetic acid	$\text{H}_3\text{C}-\text{COOH}$	
4	Butyric acid	$\text{H}_3\text{C}-(\text{CH}_2)_2-\text{COOH}$	
14	Myristic acid	$\text{H}_3\text{C}-(\text{CH}_2)_{12}-\text{COOH}$	
16	Palmitic acid	$\text{H}_3\text{C}-(\text{CH}_2)_{14}-\text{COOH}$	
18	Stearic acid	$\text{H}_3\text{C}-(\text{CH}_2)_{16}-\text{COOH}$	
20	Arachidic acid	$\text{H}_3\text{C}-(\text{CH}_2)_{18}-\text{COOH}$	
22	Behenic acid	$\text{H}_3\text{C}-(\text{CH}_2)_{20}-\text{COOH}$	
24	Lignoceric acid	$\text{H}_3\text{C}-(\text{CH}_2)_{22}-\text{COOH}$	

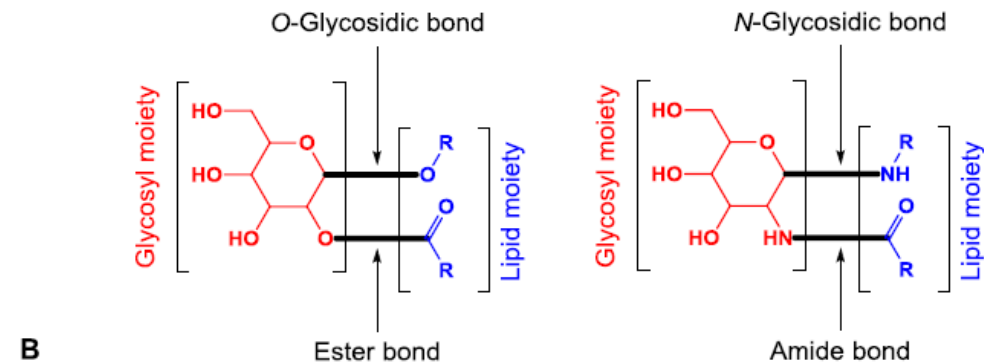
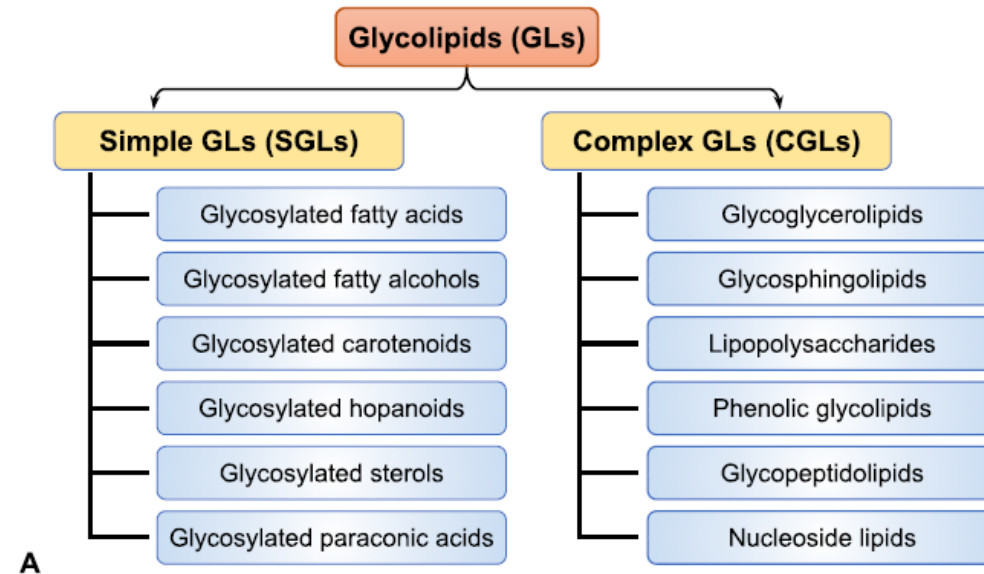
Phospholipids

- 2% of dietary lipids
- Found in foods -plant & animal origin
- Body can synthesize
- Soluble in both water & fat (polar lipids)
- Components of cell membrane
 - Blood/body fluids
 - Head – hydrophilic > Phosphate group
 - Tail – hydrophobic > FA group



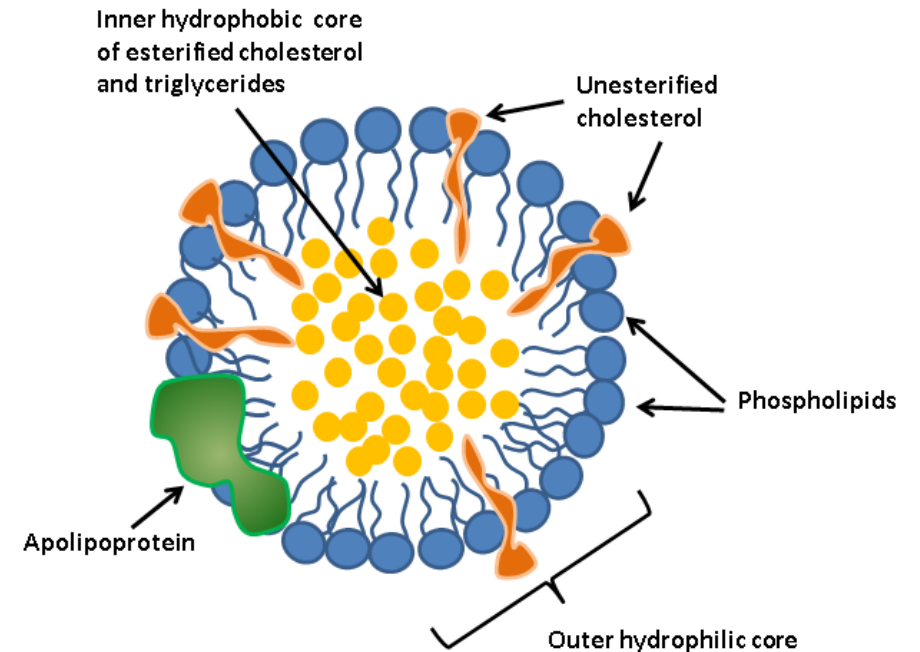
Glycolipids

- Lipids with a carbohydrate attached by a glycosidic (covalent) bond.
- Their role is to maintain the stability of the cell membrane and to facilitate cellular recognition

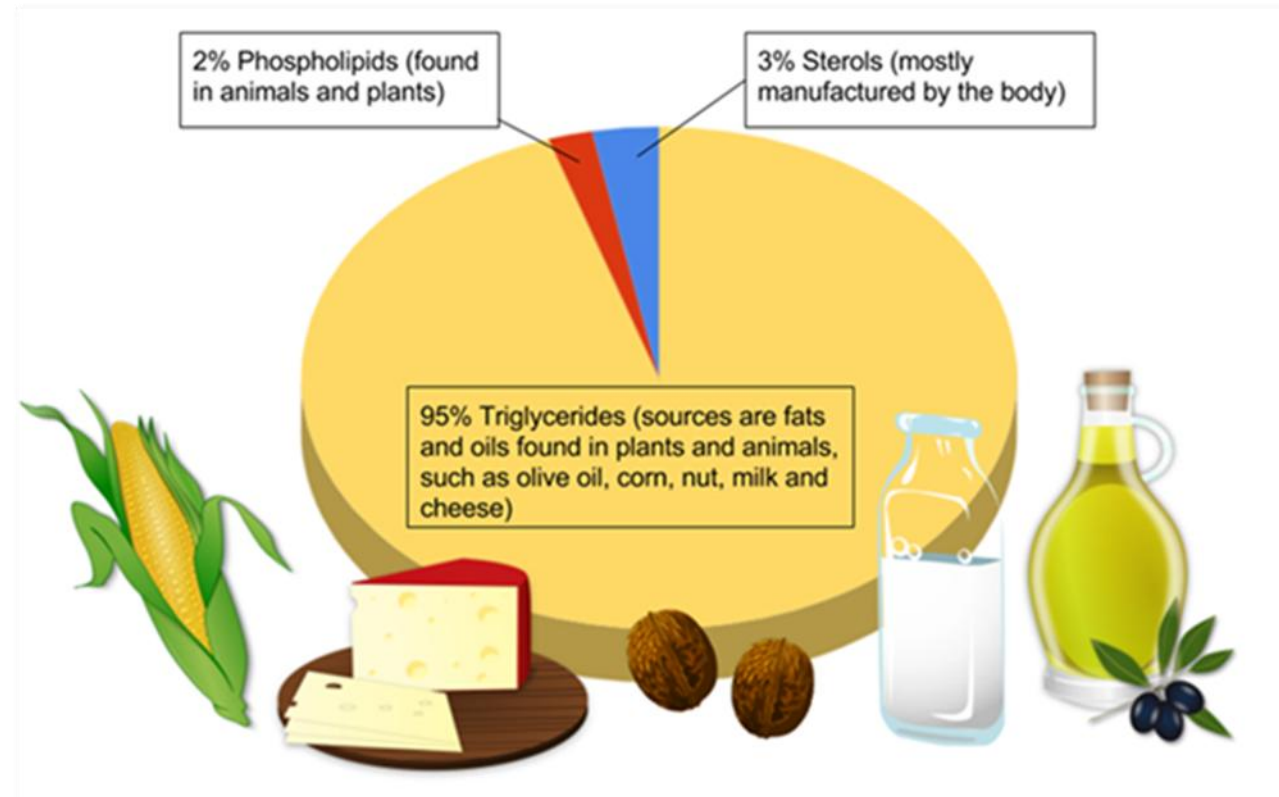


Lipoproteins

- Lipoproteins are particles made of protein and fats (lipids)
- They carry cholesterol through bloodstream to cells.
- The two main groups of lipoproteins are called **HDL (high-density lipoprotein)** or "good" cholesterol and **LDL (low-density lipoprotein)** or "bad" cholesterol

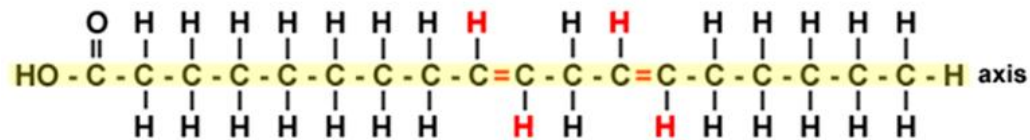


Fats in our diet

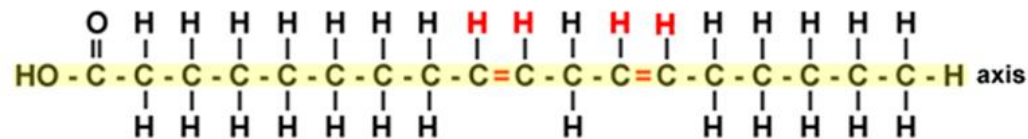


Essential and Non-essential Fatty acids

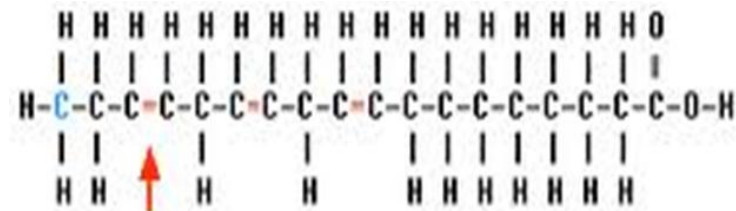
- Fats that cannot synthesis in the body to meet the physiological demand of the body are called EFAs.
- Long chain FAs
- Those should be consumed through the diet.
- About 3% of the diet should consist of EFA
- Ex : Linoleic (ω 6) and Lenolenic (ω 3) – 18 C



linoleic acid: *trans* configuration (*trans* isomer)



linoleic acid: *cis* configuration (*cis* isomer)

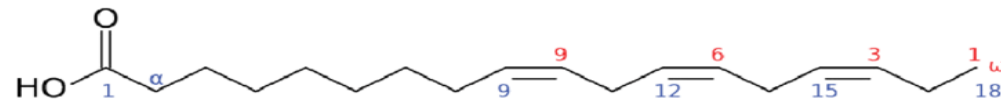
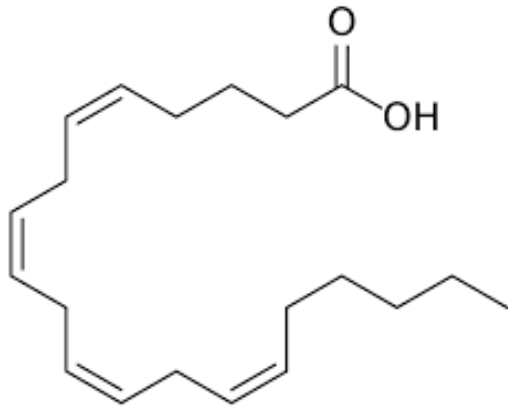


linolenic acid, an omega-3 fatty acid
(the omega carbon atom is shown in blue)

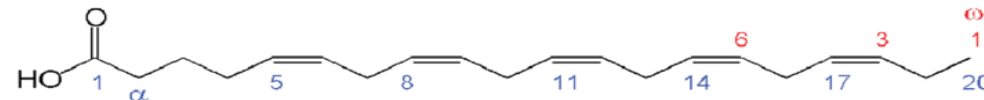
Essential FA cont..

- Arachidonic – 20 C, 4 double bonds (ω 6) : product of linoleic acids, become essential when linoleic acids are not available in the diet.
- Eicosapentaenoic (EPA) – 20 C, 5 double bonds
- Docosahexaenoic (DHA) – 22 C, 6 double bonds
- Both are ω 3 FAs
- Both EPA & DHA come from cold water fatty fish

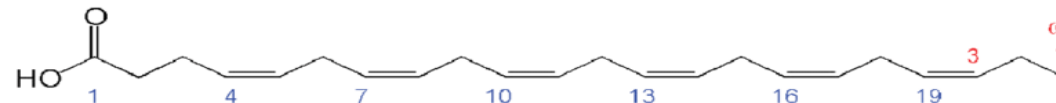
Arachidonic acid



a. α -linolenic acid (18:3, n-3; ALA)



b. Eicosapentaenoic acid (20:5, n-3; EPA)



c. Docosahexaenoic acid (22:6, n-3; DHA)

Food sources of ω 3 FA

- Fish – oily fish (Tuna, salmon, mackerel, herring)
- Sea foods
- Vegetable oils
- Nuts (especially walnuts)
- Flax seeds oil
- Leafy vegetables



Food sources of ω 6 FA

- Seed oil
- Avocado
- Soya
- Corn
- Sunflower oil
- Meat
- Fish
- Eggs



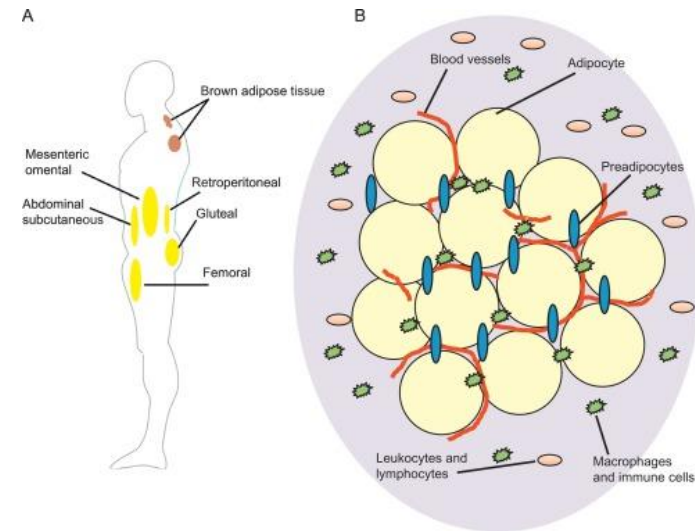
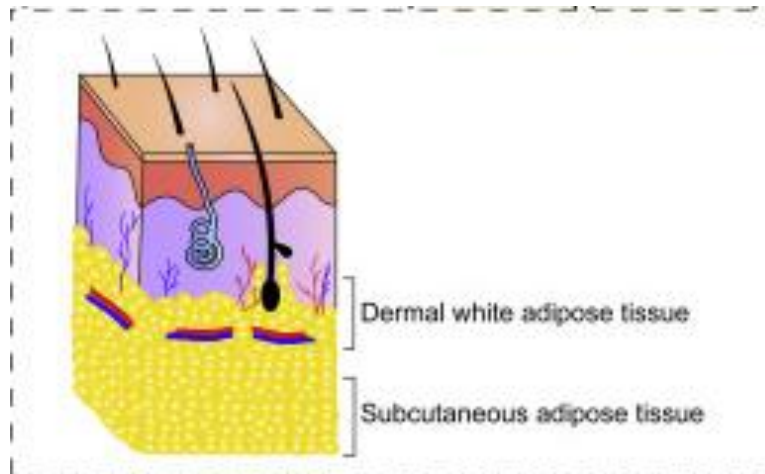
Functions of total fats

1. Energy source

- In adult : < 30 % of energy should come from fat
- In infants : 55% of energy from fats in breast milk
- Important source of energy for heart

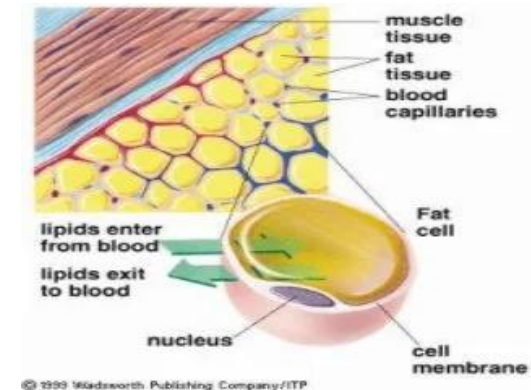
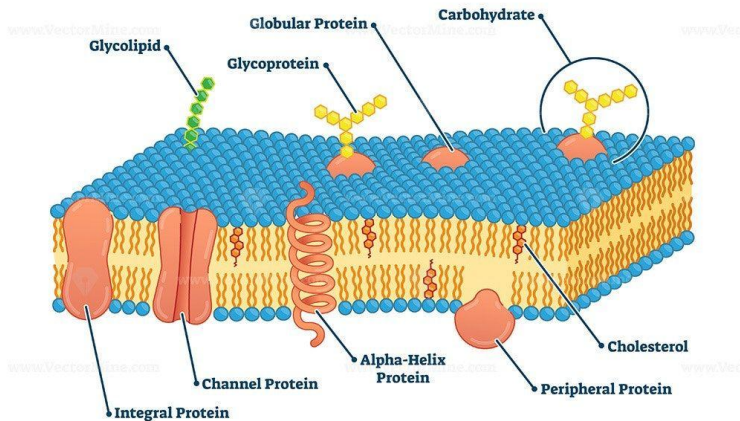
2. Storage form of energy : in adipose tissues in fat cells

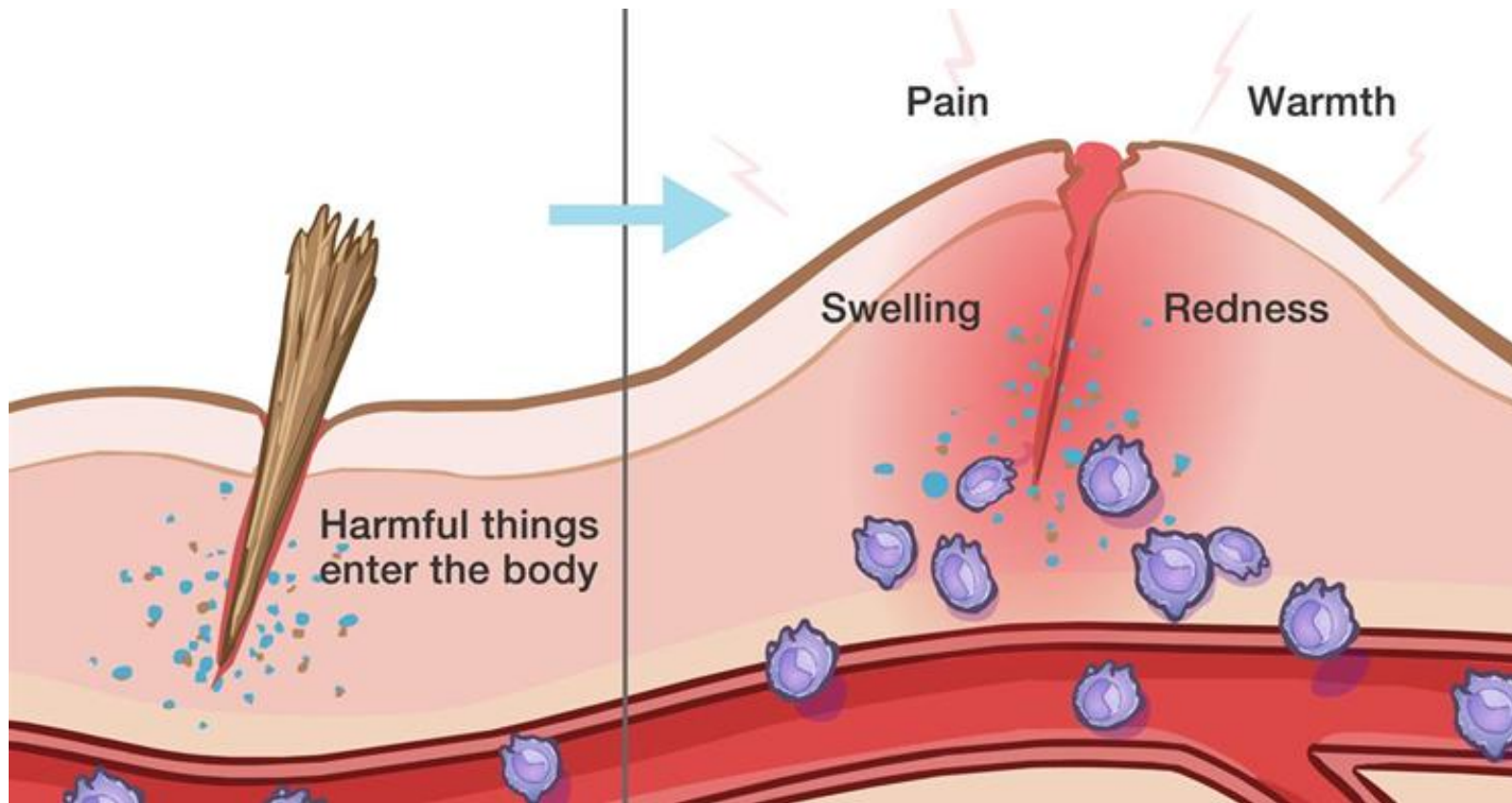
3. For healthy skin : maintain cell permeability



Functions of total fats

4. Transports cholesterol around the body
5. Precursor to prostaglandins - chemicals regulating the inflammation process
6. Structural component : cell membrane, nerves, retina, reproductive organs
7. Steroid hormones (estrogens, testosterone)
8. Insulation material to protect body from cold





Functions of fats cont...

9. Source for fat soluble vitamins

10. Increased satiety (a sense of fullness and satisfaction after eating)

Functions of EFAs

- Maintain growth, skin health, liver function, kidney functions, reproductive health
- Omega 6 – Structural molecule in skin
- Deficiency cause skin pigmentation, excess sebum secretion, increase skin permeability
- Omega 3 – Visual functions and nerves functions
- Precursor of Eicosanoids - Hormone like chemical messenger
- Role in inflammation (reactions in injury and infections)
- Maintain blood pressure
- Immune functions

Functions of EFAs cont...

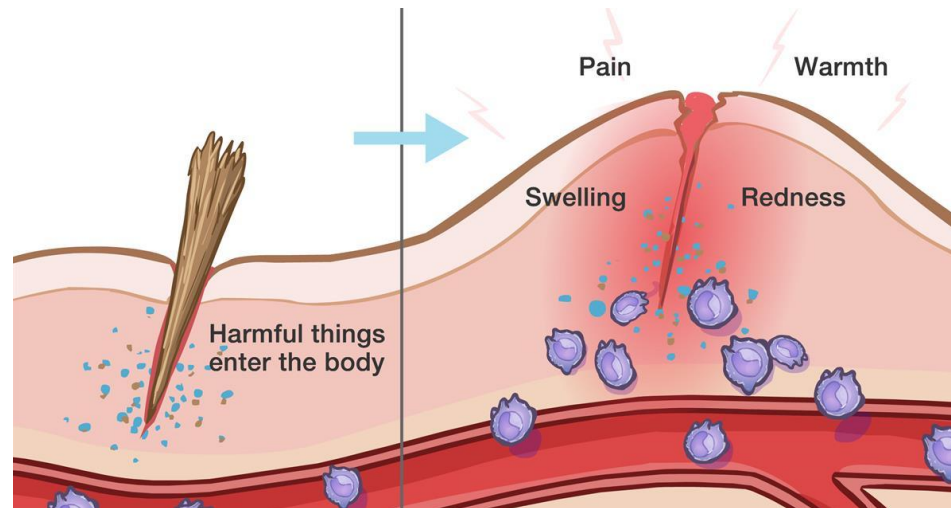
- Fetal and infants growth and development
- Mainly for brain and vision activity

EPA & DHA

- Maintain heart health
- Reduce TG level in blood
- Maintain reproductive health : Reduce pre-term deliveries
- Reduce cancer risk
- Normal eye and brain developments in babies
- Maintain mental health.

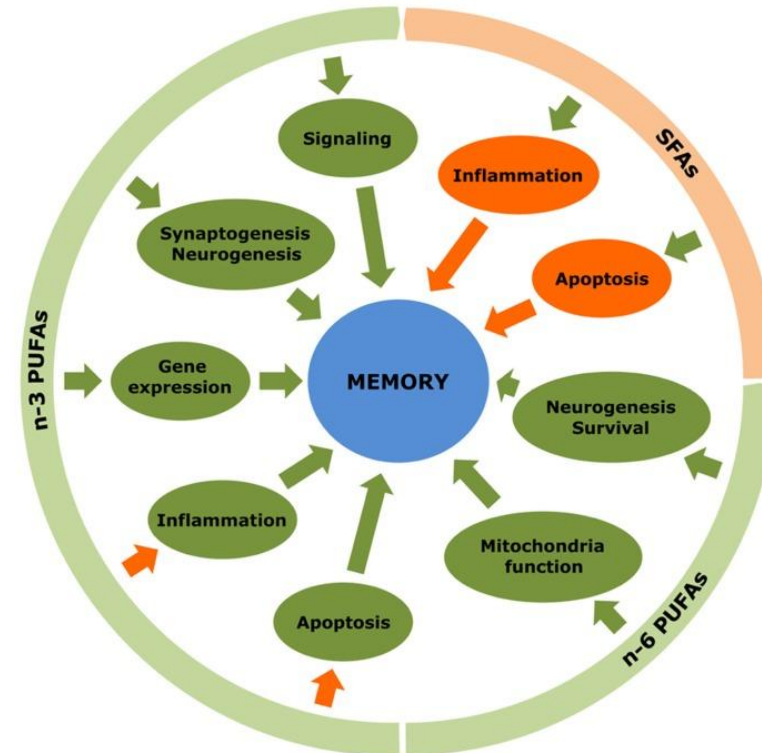
Functions of MUFA

- Reduce bad cholesterol (LDL)
- Reduce the risk of heart diseases and stroke
- Anti-inflammatory effect (inflammation means the reactions such as redness, swelling, and pain)
- Cell development
- Provide vitamin E



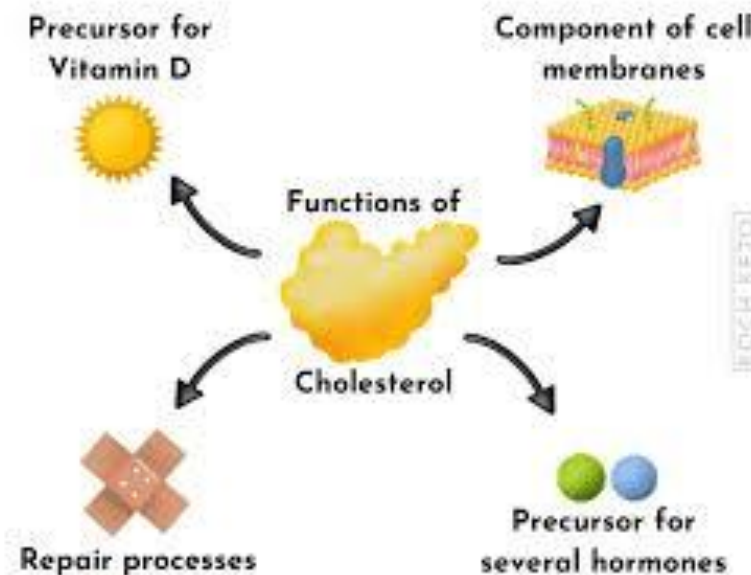
Functions of PUFA

- Act as precursors for messengers (eicosanoids, docosanoids).
- Thereby, they regulate inflammation, immunity, blood vessels, platelets, cellular growth, pain, sleep, etc
- Prevent insulin resistance/diabetes
- Reduce LDL
- Stabilize heart rhythms



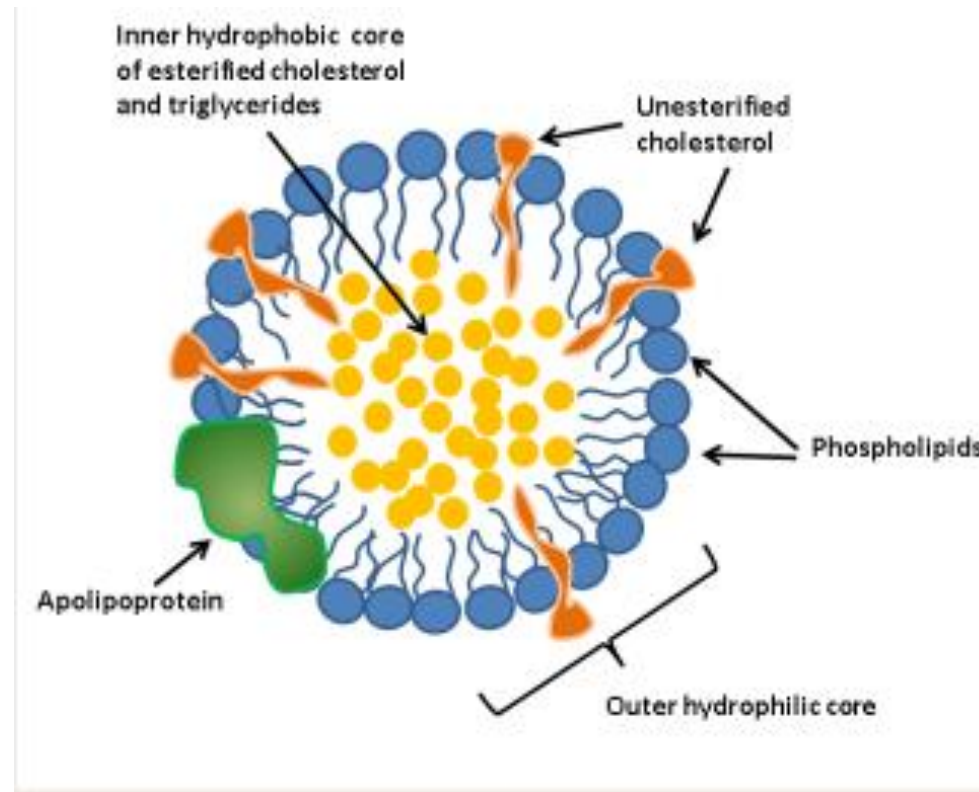
Functions of Cholesterol

- Formation of bile
- Component in cell membrane
- Precursor for vitamin D
- Precursor for steroid hormones
- Repair body tissues



Functions of Lipoproteins

- Transport TG from liver to other tissues
- Absorb fat from diet
- Remove excess cholesterol from tissues



Digestion and absorption of fats

- <https://www.youtube.com/watch?v=NoupEo5eB5U>
- <https://www.youtube.com/watch?v=B2rOqe8e75E>
- <https://www.youtube.com/watch?v=PQcJRvbBI8M>
- Melting of fats using salivary lipase is the 1st step
- Triglycerides > Diglycerides (by salivary lipase)
- Fat separate in stomach into fat globules
- Enter into small intestine (SI) slowly
- Fats are emulsified in SI, using bile

- Triglycerides digest into monoglycerides and free fatty acids

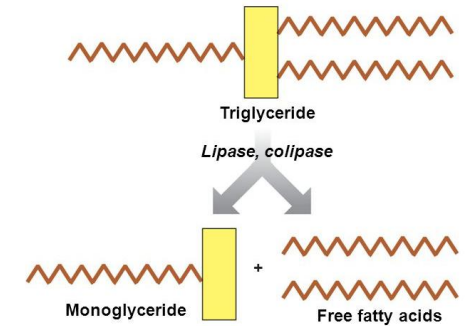
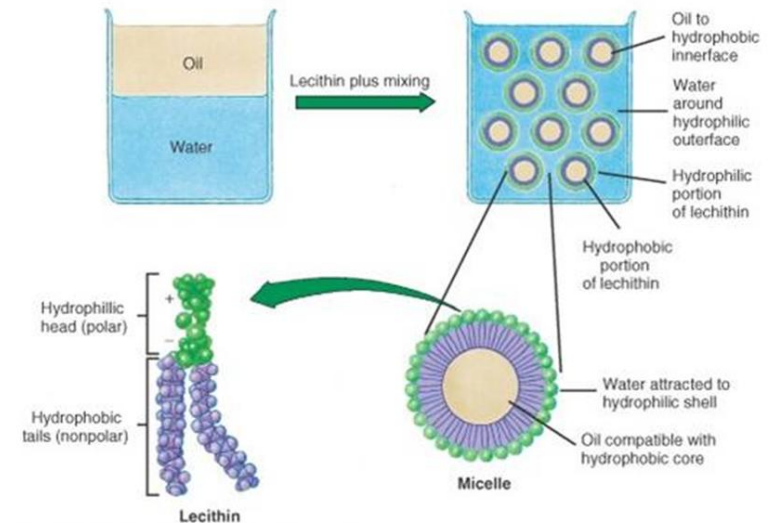
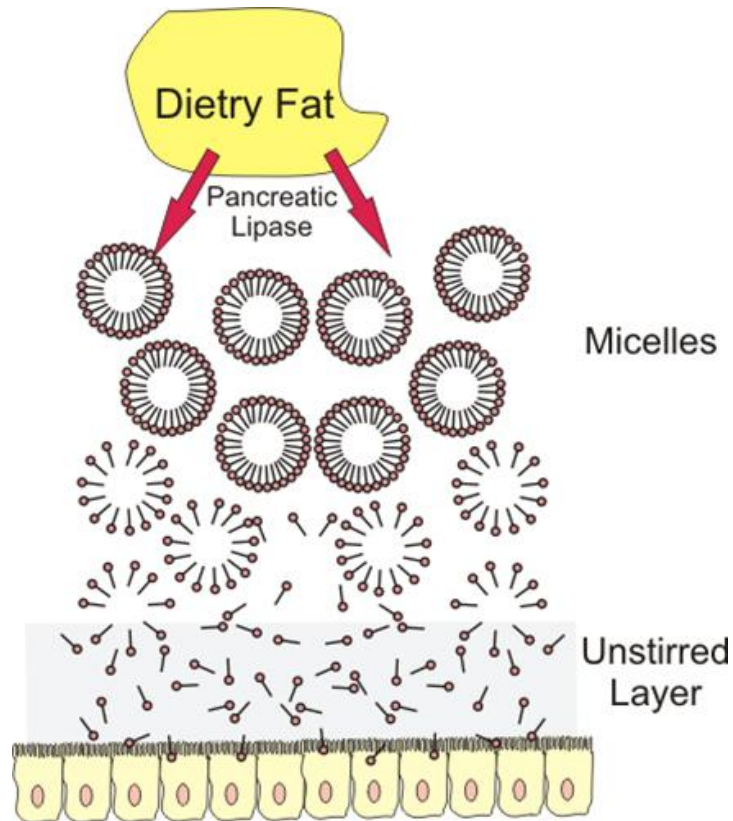


Figure 21-18



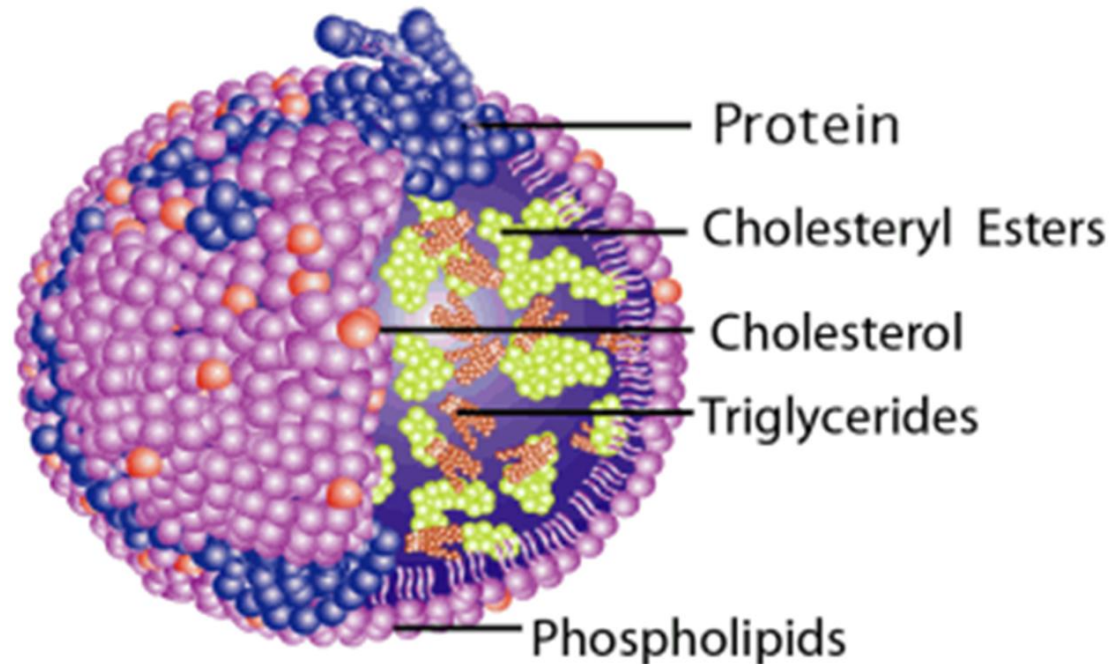
Digestion and absorption of fats

- Fats $\rightarrow$ glycerol, monoglycerides, FA (by pancreatic lipase) in small intestine
- Formation of micelles & migrate into villi

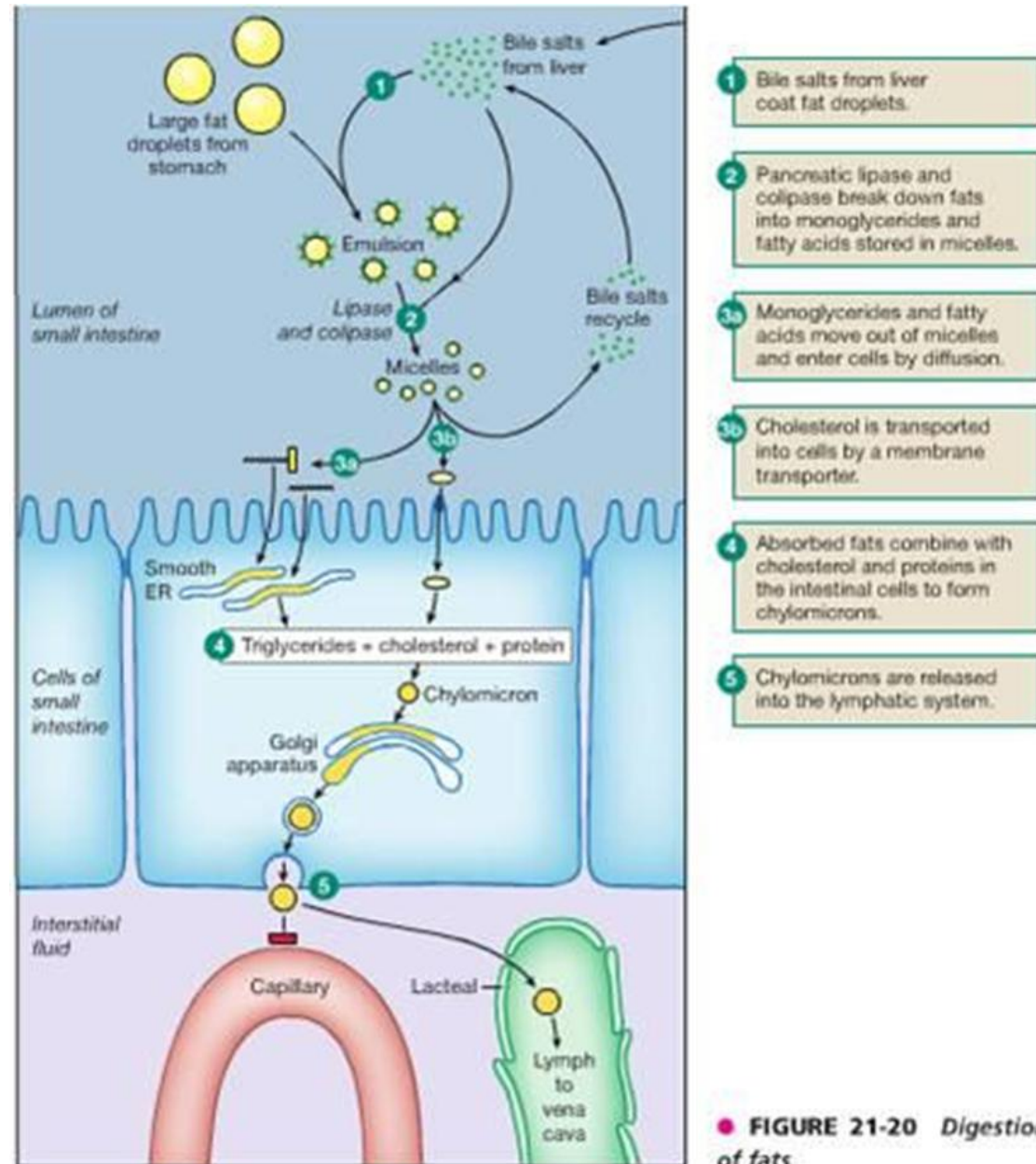


Digestion and absorption of fats

- In SI micro villi, **triglycerides are resynthesized** and are packed into chylomicrons (Lipoproteins)
- These enter to lymph
- Small lipid fragments (small FFAs) enter to blood



Digestion and absorption of fats



● **FIGURE 21-20** Digestion of fats

Transportation of fats

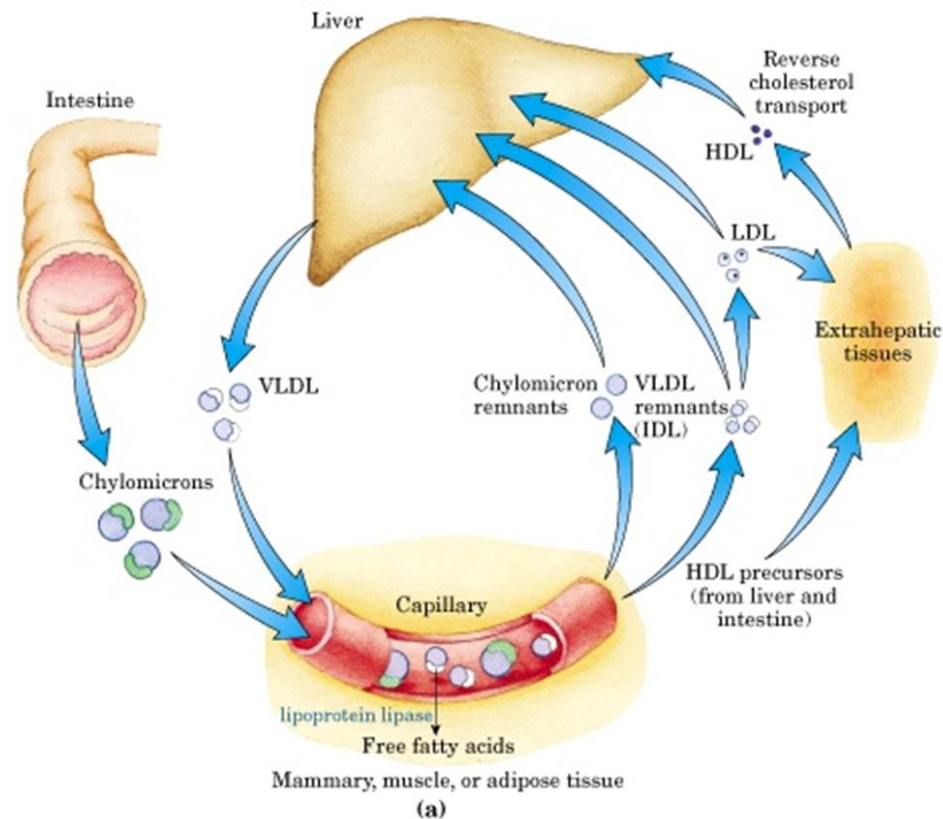
- While chylomicrons transport in circulation FA are released to cells
- Some FFAs used for energy production
- Some FA travel through the blood stream bound to albumin, and they are stored in muscle as triglyceride
- Rest of fats (TG, cholesterol, phospholipids) travel to liver to reprocess
- In liver, lipids are repacked with new protein layer (protein surrounded lipids)
- Call VLDL (very low density lipoproteins)

Transportation of fats

- VLDL enters into circulation, release their lipids to the circulation and become LDL (Low density lipoproteins)
- HDL (High density lipoproteins) are formed in liver and Intestine using LDL and HDL precursor

Transportation of fats cont...

- LDL risk factor for heart diseases (bad fats)
- HDL – It carries cholesterol, glycerol, FA released from tissues, back to liver for processing (good fats)



FFAs in body

- FA travel through the blood stream bound to albumin
- Contribute to energy requirements of exercise
- During moderate exercise, about 1/2 of the calories are from fat ; after about 1 hour of exercise, there is a gradual increase in the use of fats for calories
- That is why we promote moderate exercise to reduce body weight

Lipid storage

- Stored in adipose tissues as triglycerides (TG)
- Lipase also available in the muscle cells
- When body needs energy TG are hydrolyzed and form FA and Glycerol
- This FA will be used to produce energy by aerobic energy production system

Health effect of total fats

- Heart diseases/stroke
- Cancer – animal fats : SFA causes for cancers
- Obesity – excess fat deposits in adipose tissues
- Diabetes – increase insulin resistant
- Fats deficiency – unable to synthesis essential hormones, visual functions, brain functions are affected

Health effects of different fat

- **LDL** – increase the risk of heart diseases (bad fats) called atherosclerosis
 - **HDL** – scavenge cholesterol from blood and LDL and travels back to liver (good fats)
 - **SFA** – increases LDL level and cause for heart diseases, should get <10% of energy from SFA
 - Present in milk and milk products, meat, butter, deep fried foods
 - **Triglycerides (TG)** : more TG bad, causes for heart diseases
 - High levels of TG contributes to hardening of the arteries (atherosclerosis) which increases risk of stroke and heart disease
- <https://www.youtube.com/watch?v=tmZksHXtZtg>

LIPID PROFILE

	DESIRABLE	BORDERLINE	HIGH RISK
Cholesterol	<200 mg/dl	200-239 mg/dl	240 mg/dl
Triglycerides	<150 mg/dl	150-199 mg/dl	200-499 mg/dl
HDL cholesterol	60 mg/dl	35-45 mg/dl	<35 mg/dl
LDL cholesterol	60-130 mg/dl	130-159 mg/dl	160-189 mg/dl
Cholesterol/ HDL ratio	4.0	5.0	6.0

Health effects of different fats

- **Trans fats** – Increases LDL and reduce HDL, over activity of the immune system, insulin resistance, increase risk of blood clotting
- Only 1% calories from trans fats are recommended
- Present in margarine (hydrogenated fats – hard margarine), fried foods, pizza, cakes, cookies, pastry
- Soft margarine has low amount



Recommended fat intake

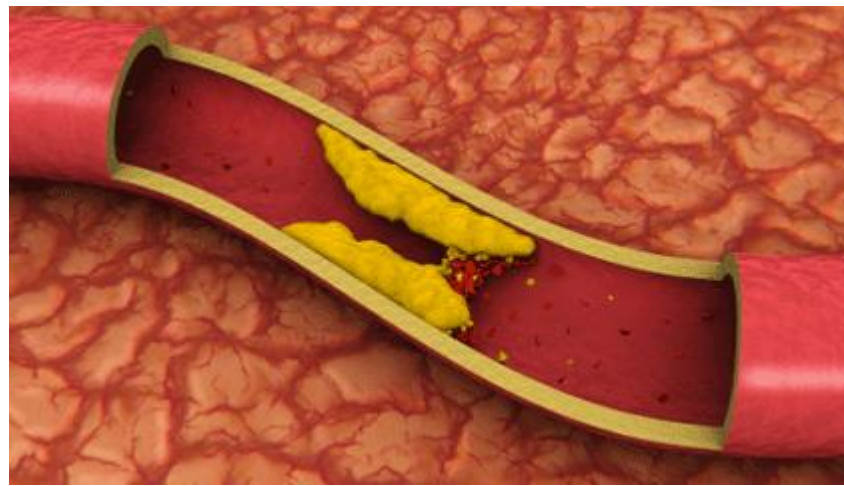
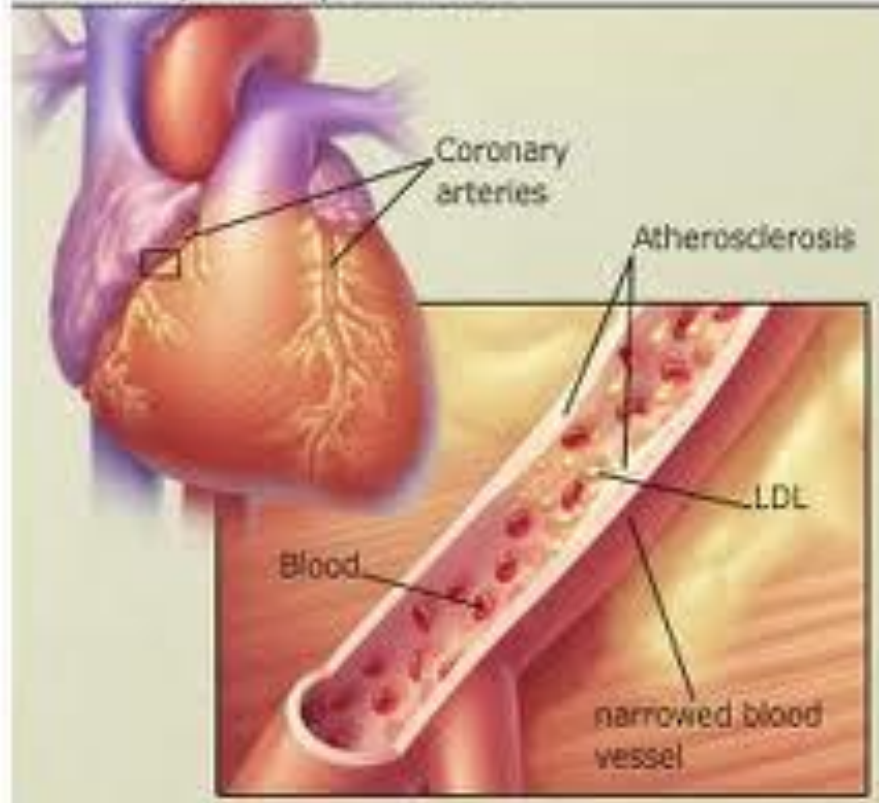
- According to the 2005 Dietary Guidelines for Americans:

- Total fat for the day should be:
- Adults: 30 % of calories
- Children: 2-3 year olds: 40 - 45% of calories
- Infants: 55%

For all of us:

- Most fat should be from poly or mono-unsaturated sources of fat (such as fish, nuts and vegetable oils)
- Limit the intake of fats and oils high in saturated and/or trans fatty acid

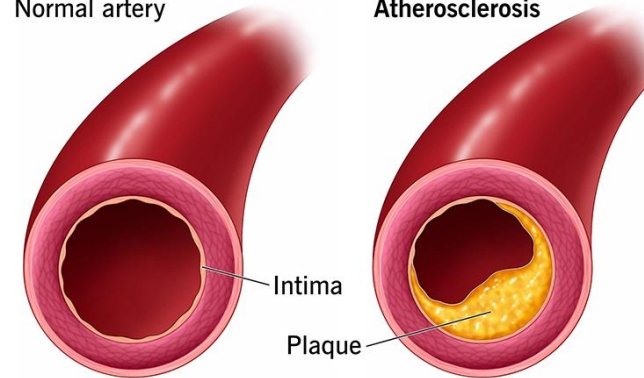
Coronary Artery Disease



Atherosclerosis

Normal artery

Atherosclerosis



Progression

